UNIFAC Parameters for Four New Groups

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Interaction parameters are presented for four new main groups added to the original UNIFAC table (Hansen, H. K.; Rasmussen, P.; Fredenslund, Aa.; Schiller, M.; Gmehling, J. *Ind. Eng. Chem. Res.* **1991**, *30*, 2352). With these new groups, it is possible to describe mixtures with aromatic nitrile groups, epoxy ethers, anhydrides, and aromatic bromo groups. In total 82 new parameter values are reported.

Introduction

Liquid-phase activities cannot be predicted in a rigorous fashion. An efficient, widely used, semiempirical method is the UNIFAC[2] group contribution method and later variants of the model.[1,3−5] The utility and efficiency of such methods depends on the availability of group volume parameters, $\boldsymbol{R}$, group surface area parameters, $\boldsymbol{Q}$, and binary group-interaction parameters, $\boldsymbol{a}$.

An extensive table of such parameters[1] was published, as the conclusion of an ongoing effort to revise and extend the UNIFAC method, in a series of five papers starting in the mid-1970s.

In later years, numerous workers[6−13] have published model parameters. However, with several different variants of the model established, different variants are employed and often the efforts are constrained to treating a single group with a single model variant. This paper attempts a unifying comprehensive treatment of new additions to the original UNIFAC model.[1]

Background

The most extensive UNIFAC parameter collection to date is probably that of Gmehling and co-workers,[3,5,12,13] which has been widely used and is now continuously revised and extended as part of an industrial/academic joint venture.[14] In fact, recently Gmehling and co-workers[12,13] indicated progress of ongoing work to extend their version to cover three of the groups treated in this paper, without giving all model parameters for full utility. Here we have considered the original[1] model. In what follows UNIFAC refers to that variant of UNIFAC. Model equations are summarized in Appendix A.

Currently, when using UNIFAC for bromo-substituted aromatic compounds, the only possible calculation is to apply the bromo group, Br, developed by fitting the model to data on aliphatic bromo compounds. However, in a number of cases, we have found that the contribution of an aliphatic bromo is not always equivalent to that of an aromatic one. Examples are the cases of bromobenzene mixtures with *p*-xylene and cumene,[15] in which false azeotropy is predicted. Also in the case of ethanol/bromobenzene data,[16] use of the aliphatic Br group predicts false liquid-phase instability. Therefore, we here develop contributions for an aromatic bromo group, AC−Br. We know of no earlier attempts to devise separate UNIFAC contributions to aromatic bromo systems.

Based on the present tables, an aromatic nitrile group cannot be constructed. Therefore, a new addition, AC−CN, seems necessary from a practical point of view. However, such developments have also frequently been necessary in order to improve the representation of data on substituted aromatics. This seems to have been generally recognized by engineers conducting research in the area of group contribution methods. For the groups to be independent of neighbors, particularly polar aromatic groups are usually considered to be different from alkyl ones. Presently, this is done with UNIFAC in the case of aliphatic and aromatic hydroxyls.

Jonasson and colleagues[6] measured phase equilibria on epoxy systems and published six interaction parameters (with three other main groups) for an epoxy main group, CH_2OCH_2. We continue their efforts by adding contributions of epoxy interactions with 14 extra main groups.

In some special cases, anhydride phase equilibria may be calculated by considering the anhydride group as a combination of the existing ester (CH_3COO) and ketone (CH_3CO) subgroups. We have done that for a number of binary mixtures including acetic anhydride, and interestingly enough reasonable phase diagrams may be obtained in a number of cases. This is noteworthy because neither the ester nor the ketone parameters were developed based upon or intended to be used for anhydride systems. However, there are also a number of mixtures for which such an ester/ketone combination does not give accurate results, and the calculations mentioned treat acetic anhydride only. Similar calculations cannot be made for important anhydrides such as maleic acid anhydride and 1,2,3,6-tetrahydrophthalic anhydride because the necessary groups are not available. Thus, there seems to be a good reason for developing an anhydride group, although much of the advantage of such a group seems to lie in its utility for ring systems. A few years ago Garcia et al.[10] gave parameters for the interaction of an anhydride group, O═COC═O, with the CH_2 main group. This was part of an investigation of the suitability of UNIFAC for treating anhydrides, but only excess enthalpies were considered. Here VLE and H^E are considered.

Thus, while some of our results may have appeared before, in other contexts, they have not yet been collected together nor put in suitable form for maximum utility. We have made no revisions of previously published group interaction parameters.

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Table 1. New Groups: Main Group Definitions and *R* and *Q* Values

main group	subgroup *k*	R_k	Q_k	sample group assignment
	CH_2OCH_2	1.5926	1.3200	ethylene oxide: CH_2OCH_2
	CH_2OCH	1.3652	1.0080	1,2-propylene oxide: CH_3, CH_2OCH
CH_2OCH_2	CH_2OC	1.1378	0.7800	1,2-epoxy-2-methylpropane: $2CH_3$, CH_2OC
	CHOCH	1.1378	0.6960	2,3-epoxybutane: $2CH_3$, CHOCH
	CHOC	0.9103	0.4680	2,3-epoxy-2-methylbutane: $3CH_3$, CHOC
	COC	0.6829	0.2400	2,3-epoxy-2,3-dimethylbutane: $4CH_3$, COC
O═COC═O	O═COC═O	1.7732	1.5200	acetic anhydride: $2CH_3$, O═COC═O
AC−CN	AC−CN	1.3342	0.9960	benzonitrile: 5ACH, AC−CN
AC−Br	AC−Br	1.3619	0.9720	bromobenzene: 5ACH, AC−Br

Parameters *R* and *Q*

Functional groups defined in this work are summarized in Table 1. The traditional approach to ***R*** and ***Q*** values[17] for use with UNIFAC models is to normalize previously published[18] van der Waals volumes, V_{wk}, and surface areas, A_{wk}, using the equations

$$R_k = V_{wk}/15.17\ \text{cm}^3\cdot\text{mol}^{-1};\qquad Q_k = A_{wk}/2.5 \times 10^9\ \text{cm}^2\cdot\text{mol}^{-1} \quad (1)$$

This approach has been taken in most cases. Appendix B describes the motivation behind ***R*** and ***Q*** values used in this work.

Parameters *a*

Careful collection and reduction of measured data are required before reliable estimates can be made. Many data, therefore, form the basis of the group interaction parameters, ***a***, reported here. In this work, the parameters have mainly been found from activity coefficient data obtained from the open literature.[19,20]

In the case of vapor−liquid data, consistency checking by means of Barker's method[21] and the area/integral test was used to assist us in evaluating the reliability of all data. In addition to vapor−liquid data, solid−liquid data, and liquid−liquid data, heats of mixing and infinite-dilution activity coefficients were occasionally included in the data reductions. In those cases graphical evaluation of the data was employed to screen out the most suspicious results.

From a liquid-phase model, for G^E, one can obtain an expression for H^E by Gibbs−Helmholtz differentiation assuming that the interaction parameters are independent of temperature. The resultant equation may then be fitted to experimental excess enthalpies to obtain explicit values for the interaction parameters that may then be used to obtain G^E. The basic premise in this procedure is, of course, that the original equation adequately predicts the temperature dependence of the system. Similarly, any liquid-phase G^E model that also describes correctly the temperature, H^E, and composition, ln γ_i^∞, derivatives of G^E, can be used to simultaneously correlate excess enthalpy, H^E, activity coefficient, and G^E data. This means that parameters estimated from H^E data can be applied for the representation of G^E and/or activity coefficient data and vice versa. In practice, however, parameter values obtained from H^E alone have not been found to be very reliable in predicting G^E. Therefore, experimental excess enthalpies have only been used in combination with sufficient numbers of activity coefficient data.

Phase equilibrium data are reduced by use of a virial equation of state[22] to model vapor nonidealities and UNIFAC to model liquid nonidealities. These equations are used in conjunction with the basic equations of phase equilibrium to calculate phase compositions from the measured data. "Optimal" parameters are determined by minimizing the following sum of squares of residuals:

$$f_{\text{VLE}} + f_{H^E} + f_\gamma + f_{\text{SLE}} + f_{\text{LLE}} + f_{\text{R}} \quad (2)$$

subject to the constraints imposed by the thermodynamic criteria of phase equilibrium. In eq 2:

$$f_{\text{VLE}} = \sum_i \left(\frac{P - \hat{P}}{\sigma_{\text{P}}}\right)_i^2 \quad (3)$$

$$f_{H^E} = \sum_i \left(\frac{H^E - \hat{H}^E}{\sigma_{\text{H}}}\right)_i^2 \quad (4)$$

$$f_\gamma = \sum_i \left(\frac{\gamma_i^\infty - \hat{\gamma}_i^\infty}{\sigma_\gamma}\right)_i^2 \quad (5)$$

$$f_{\text{SLE}} = \sum_i \left(\frac{x_{21} - \hat{x}_{21}}{\sigma_{\text{x}}}\right)_i^2 \quad (6)$$

$$f_{\text{LLE}} = \sum_i \left(\frac{x_{21} - \hat{x}_{21}}{\sigma_{\text{x}}}\right)_i^2 \quad (7)$$

In all eqs 3−7 a circumflex denotes a calculated value, based on the parameters obtained from data reduction. P is total pressure, and x_{21} is the concentration of solute 2 (solid−liquid) in solvent 1. i runs over the respective number of data points, i.e., VLE data points in eq 3, excess enthalpy data points in eq 4 and so forth. The last term in eq 2 is

$$f_{\text{R}} = \frac{\alpha}{\delta}\sum_m\sum_n (a_{mn} - a_{mn}^0)^2 \quad (8)$$

where the sum over m and n is over all binary interaction parameter pairs. This is called a regularization term.[23] Previously, such terms were employed in regression of LLE data.[24] We have also found such terms quite useful in other situations and have, therefore, used them in all cases mentioned above. α was set to the ratio between the total number of data points divided by the number of parameters estimated. Typically, $\delta = 10^5$ and $a_{mn}^0 = 0$ were employed.

A single residual—one involving the error on the total pressure P (but none involving the error on y_1)—is introduced per binary VLE data point in eq 3. Van Ness et al.[25] compared, in detail, advantages and disadvantages associated with a number of different objective functions for reduction of binary VLE data. Their

Table 2. Objective Function, Equation 2, Weighting Factors

quantity X	σ_x	quantity X	σ_x
P	$0.01P$	γ	$0.1\gamma_i^\infty$
H	$0.05H^E$	x	0.01

Table 3. Interaction Parameters: AC−Br

i	j	a_{ij}/K	a_{ji}/K
AC−Br	CH_2	−20.31	153.72
AC−Br	ACH	−106.74	174.35
AC−Br	$ACCH_2$	568.47	−280.90
AC−Br	OH	284.28	147.97
AC−Br	H_2O	401.20	580.28
AC−Br	CH_2CO	106.21	179.74
AC−Br	CCl_4	−108.37	127.16
AC−Br	ACCl	5.76	8.48
AC−Br	$ACNO_2$	−272.01	1742.53
AC−Br	ACF	107.84	117.59
AC−Br	DMF	−33.93	39.84

Table 4. Interaction Parameters: CH_2OCH_2

i	j	a_{ij}/K	a_{ji}/K
CH_2OCH_2	CH_2	21.49[a]	408.30[a]
CH_2OCH_2	C═C	−2.80[a]	219.90[a]
CH_2OCH_2	ACH	344.42	171.49
CH_2OCH_2	$ACCH_2$	510.32	−184.68
CH_2OCH_2	OH	244.67	6.39
CH_2OCH_2	CH_3OH	163.76	98.20
CH_2OCH_2	H_2O	833.21	−144.77
CH_2OCH_2	CH_2CO	569.18	−288.94
CH_2OCH_2	CHO	−1.25	79.71
CH_2OCH_2	CH_2COO	−38.40	36.34
CH_2OCH_2	HCOO	69.70	−77.96
CH_2OCH_2	CH_2O	−375.60	567.00
CH_2OCH_2	COOH	600.78	12.55
CH_2OCH_2	CCl	291.10[a]	−127.90[a]
CH_2OCH_2	CCl_3	−286.26	165.67
CH_2OCH_2	CCl_4	−52.93	291.87
CH_2OCH_2	Cl−(C═C)	177.12	−127.06

[a] Previously published.[6]

Table 5. Interaction Parameters: O═COC═O

i	j	a_{ij}/K	a_{ji}/K
O═COC═O	CH_2	272.82	718.01
O═COC═O	C═C	569.71	−677.25
O═COC═O	ACH	165.18	272.33
O═COC═O	$ACCH_2$	369.89	9.63
O═COC═O	CH_2CO	−62.02	91.01
O═COC═O	CH_2COO	−229.01	446.90
O═COC═O	CH_2O	−196.59	102.21
O═COC═O	pyridine	100.25	98.82
O═COC═O	COOH	472.04	−60.07
O═COC═O	CCl_4	196.73	532.73
O═COC═O	CS_2	434.32	684.78
O═COC═O	I	313.14	190.81
O═COC═O	COO	−244.59	−100.53

Table 6. Interaction Parameters: AC−CN

i	j	a_{ij}/K	a_{ji}/K
AC−CN	ACH	920.49	22.06
AC−CN	$ACCH_2$	305.77	795.38
AC−CN	COOH	171.94	88.09

conclusions are still valid:[26] Although reduction of isothermal VLE data can be accomplished well by several methods, the unweighted least-squares technique minimizing the P residuals is at least as good as any and is certainly the simplest and most direct. In using the objective function of eq 3, we do assign weights to the P residuals, though, as given by the σ values summarized in Table 2. The interaction parameters resulting from data reduction are given in Tables 3−6.

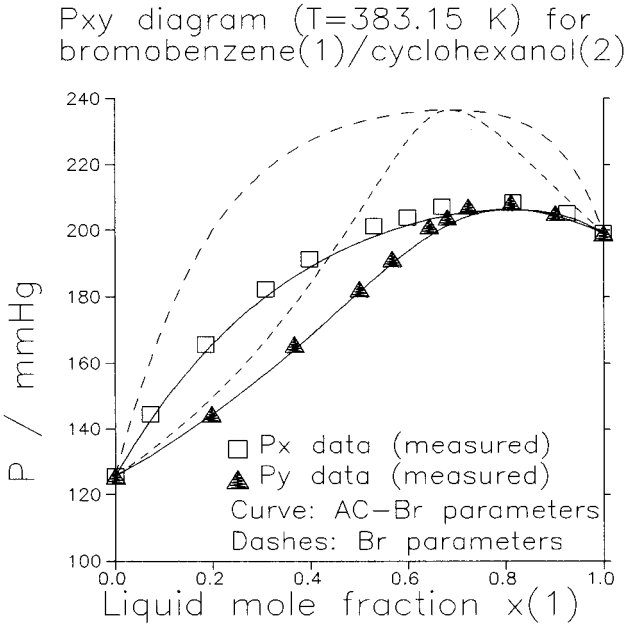


Figure 1. Pxy diagram[27] calculations with contributions of aliphatic, Br, and aromatic, AC−Br, bromo groups.

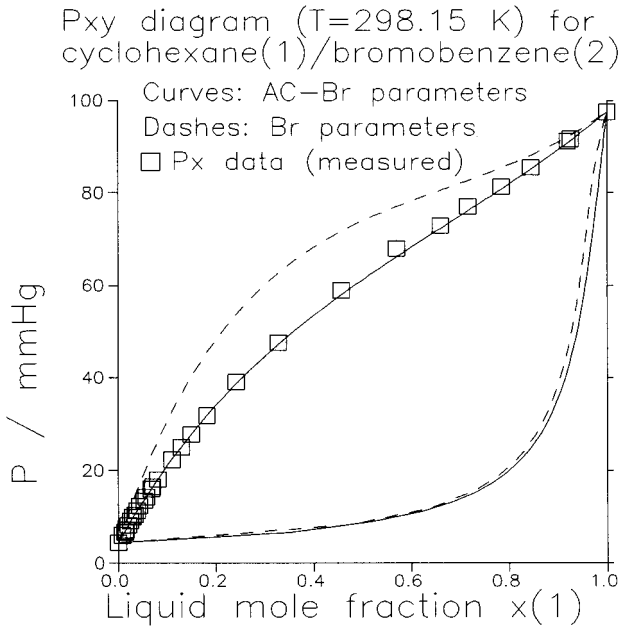


Figure 2. Pxy diagram[28] calculations with contributions of aliphatic, Br, and aromatic, AC−Br, bromo groups.

Results

Numerous results could be shown using the new parameter tables. Here we show a few representative results and some noteworthy results for polyfunctional systems.

Figures 1 and 2 indicate the improvement over the original Br parameters that is found with the aromatic contributions of Table 3. Other examples involving bromobenzene are given by Figure 3 for butanol with bromobenzene and by Figure 4 for fluorobenzene with bromobenzene, for which agreement with measurement is good. In the case of hydrocarbon/bromobenzene binaries, reliable infinite-dilution activity coefficients are available for comparison. Figure 5 shows a comparison between infinite-dilution activity coefficients calculated with the new parameters for normal alkanes

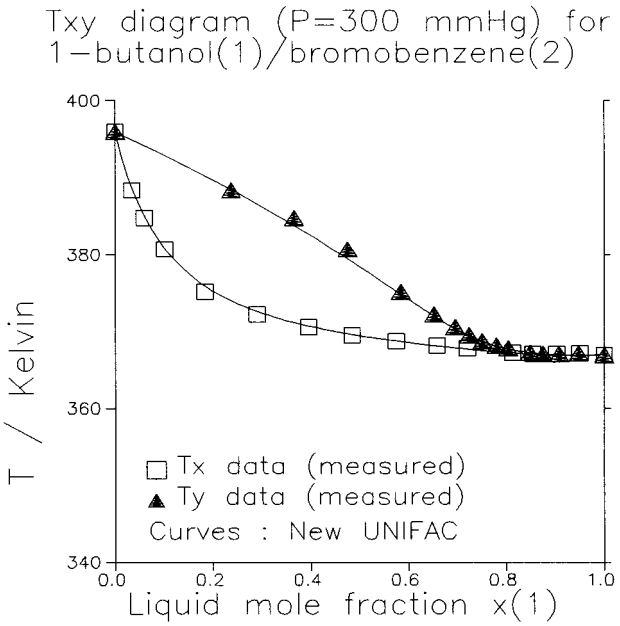


Figure 3. Txy diagram[29] calculation with contributions of an aromatic, AC−Br, bromo group.

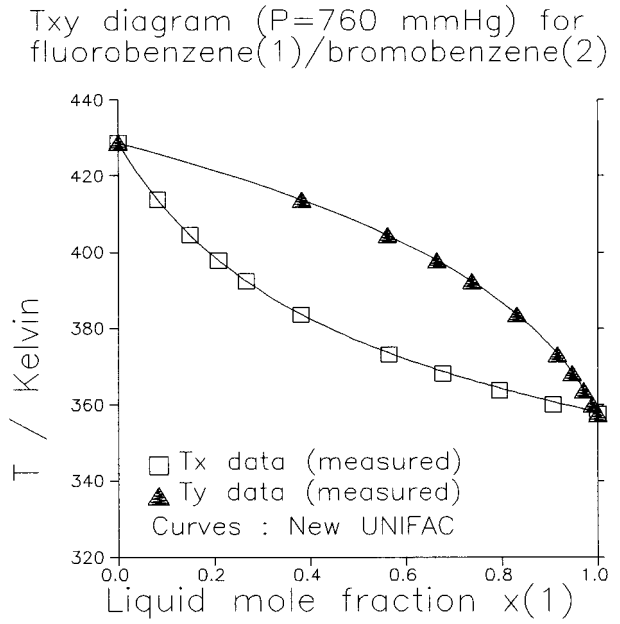


Figure 4. Txy diagram[30] calculation with contributions of an aromatic, AC−Br, bromo group.

in bromobenzene and measured data.[31] Table 7 provides additional examples for branched hydrocarbon/bromobenzene data.

In the case of dibromobenzenes, the predictions are the same for ortho, meta, and para substitution. Therefore, if the experimental data differ significantly for the three forms, there will also be significantly different errors. The majority of reliable phase equilibrium data on aromatic bromo systems are for monosubstituted substances, and these are the types of systems the model parameters have been fitted to. We have made some calculations on dibromobenzenes though. Figure 6 shows predictions of *p*-dichlorobenzene (solid) solubilities in *p*-dibromobenzene,[32] in the case of which the agreement is good. Such mixtures are almost ideal, just as mixtures of monochloro- and monobromobenzene[30]

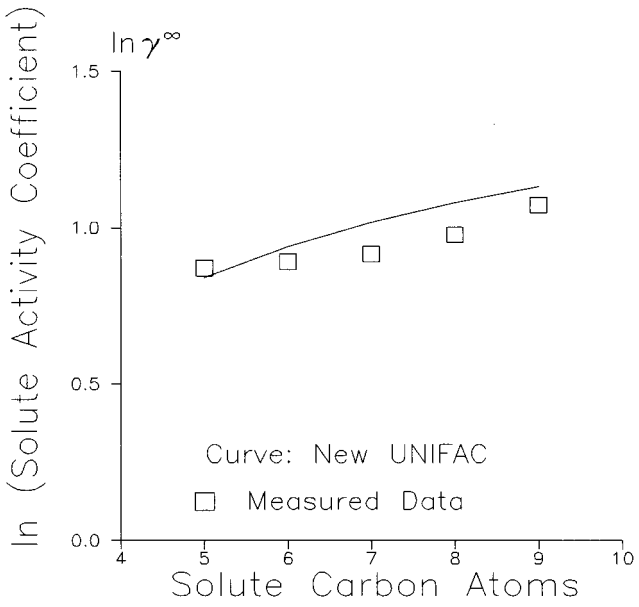


Figure 5. Infinite-dilution activity coefficients of *n*-alkanes in bromobenzene[31] at 298.15 K. Calculations with contributions of an aromatic, AC−Br, bromo group.

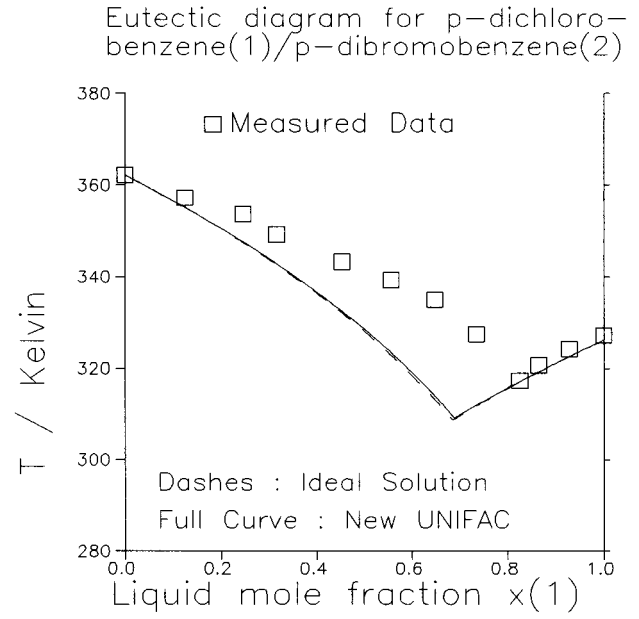


Figure 6. Eutectic diagram for *p*-dichlorobenzene (1)/*p*-dibromobenzene (2).[32] Calculations with contributions of an aromatic, AC−Br, bromo group.

Table 7. Infinite-Dilution Activity Coefficients[31] (298.15 K)

solvent (1)/solute (2) pair	γ_2^∞ measd	γ_2^∞ calcd	% error
bromobenzene/cyclohexane	1.86	1.97	−5.9
bromobenzene/2-methylpentane	2.66	2.55	4.1
bromobenzene/2,4-dimethylpentane	2.91	2.75	5.4
bromobenzene/2,3,4-trimethylpentane	2.69	2.93	−8.8
bromobenzene/ethylcyclohexane	2.14	2.35	−9.7
n-octadecane/bromobenzene	1.34	1.36	−1.6

are. Predictions, however, do not agree as well with measured *p*-dibromobenzene (solid again) solubilities in *p*-dichlorobenzene, as shown in Figure 6. To fit these data, surprisingly large activity coefficients are required over the entire composition range. This is in contrast

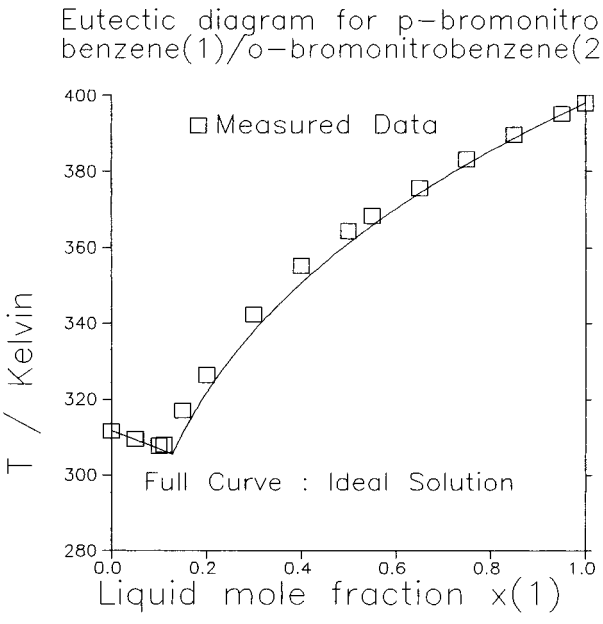


Figure 7. Eutectic diagram for *p*-bromonitrobenzene (1)/*o*-bromonitrobenzene (2).[33]

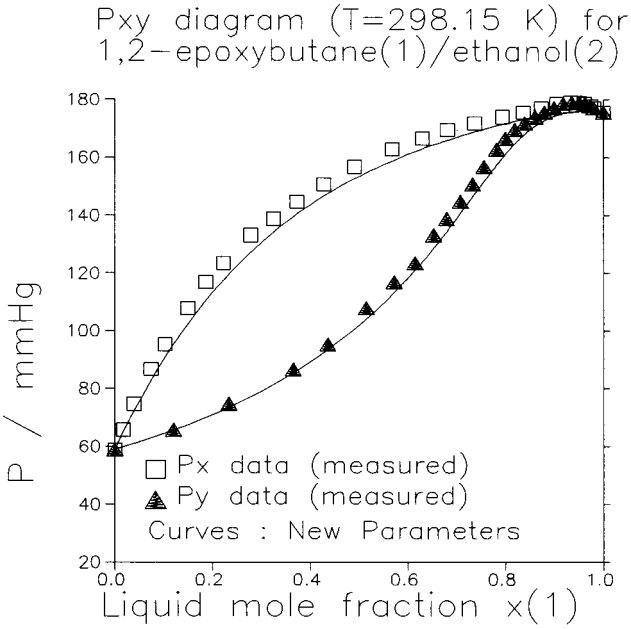


Figure 8. Pxy diagram[35] calculation with contributions of new CH_2OCH_2 parameters.

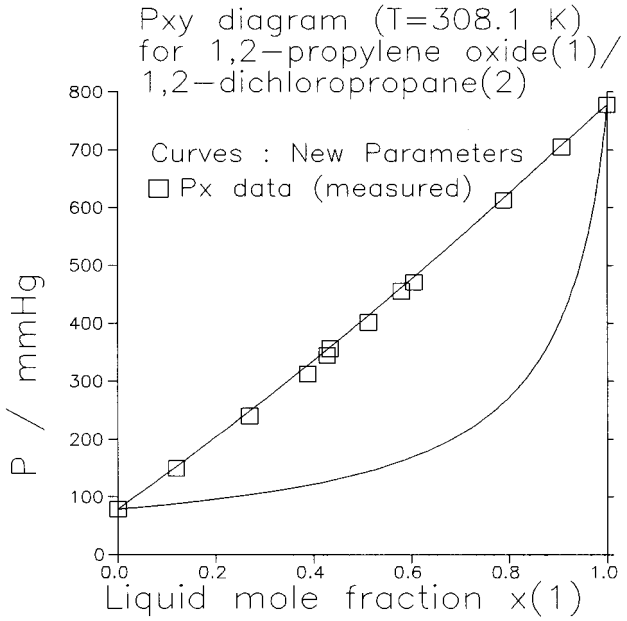


Figure 9. Pxy diagram[36] calculation using CH_2OCH_2 parameters.

to what the model says and in contrast to the trends in other data we encountered on similar mixtures. In addition to multiple bromosubstitution, the general effect of ortho, meta, and para isomerism for substituted bromobenzenes can be important. There are some fairly old VLE and SLE data[33] on systematic binary combinations of *o*-, *m*-, and *p*-bromonitrobenzene. The case of the ortho plus para binary is given in Figure 7. In this case, a solution-of-groups model predicts ideal solution behavior, as shown. This is in fair agreement with the data. Good agreement in this case seems to indicate that the collections of atoms that have been formed into groups work effectively for this system. Conceivably, nonideality should result from interfering multiple substitution on the ortho form; however, this effect is not significant in this case. Other data on ortho-substituted bromobenzenes[32,34] exist. These were left out because some of the necessary interaction parameters were missing. Yet, although the cases tested here give reasonable results, the full utility of the parameters in the cases of multiple substitution is not yet fully known.

With parameters in Table 4, Figures 8−10 plot phase diagrams for the binaries:

1,2-epoxybutane/ethanol (298.15 K)[35] (I)

1,2-propylene oxide/1,2-dichloropropane (308.1 K)[36] (II)

2,3-epoxy-1-propanol/α-epichlorohydrin (323.15 K)[37] (III)

Here system (I) involves the interactions of CH_2OCH_2 with CH_2 and OH, while (II) involves CH_2OCH_2 with CH_2 and CCl. Finally, (III) involves CH_2OCH_2 with CH_2, OH, and CCl at the same time. Yet, agreement is quite good. System (III) involves the proximity effect of a hydroxyl group situated on the same carbon atom as the epoxy group. In the case shown, such effects do not seem strong enough to cause problems. In fact, the parameters predicting azeotropy in the case of system (I) correctly predict ideality in the case of system (II).

Keeping previously published parameters fixed creates a problem when representing mixtures on epoxides and other ethers. For example, for the three binaries

MTBE/1,2-propylene oxide (i)

1,4-dioxane/α-epichlorohydrin (ii)

tetrahydrofuran/α-epichlorohydrin (iii)

If the ether/epoxy interaction is set to zero, the method predicts less nonideality than experimental data for set (i), whereas a false minimum temperature azeotrope is predicted for set (ii), indicating that the model yields a too strong positive deviation. Similarly in set (iii) the estimated deviation is too strongly positive compared with the experimental data. Thus, for representing set (i), the effect of the ether/epoxy interaction

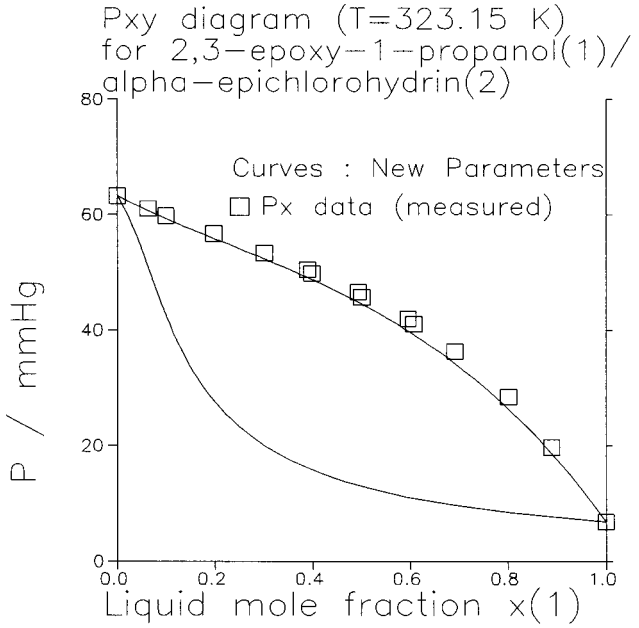


Figure 10. Pxy diagram[37] calculation using new CH_2OCH_2 parameters.

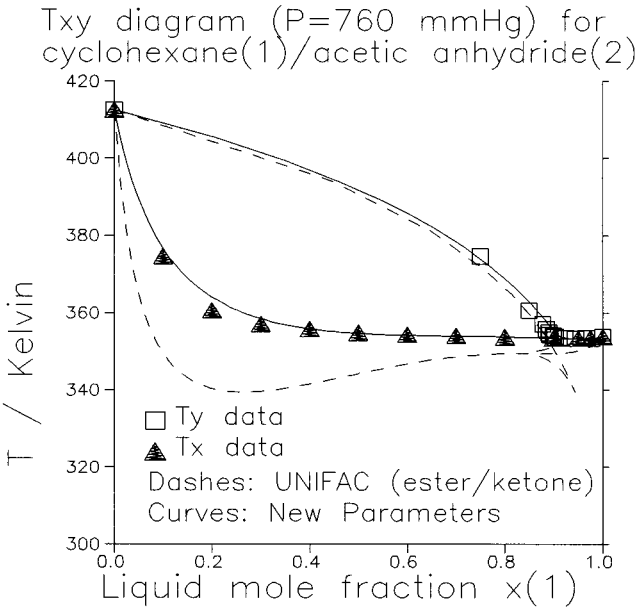


Figure 11. Txy diagram[38] calculated with an ester/ketone combination and with new parameters.

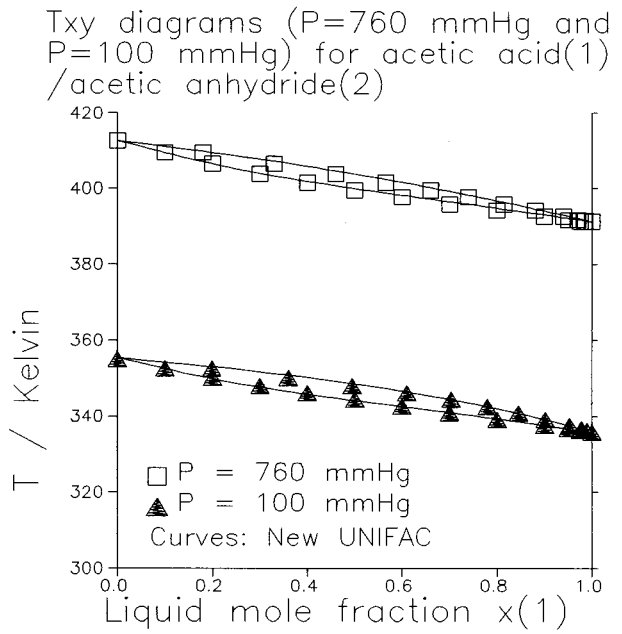


Figure 12. Txy diagrams[38] calculated with new anhydride parameters. Compare with Figure 13.

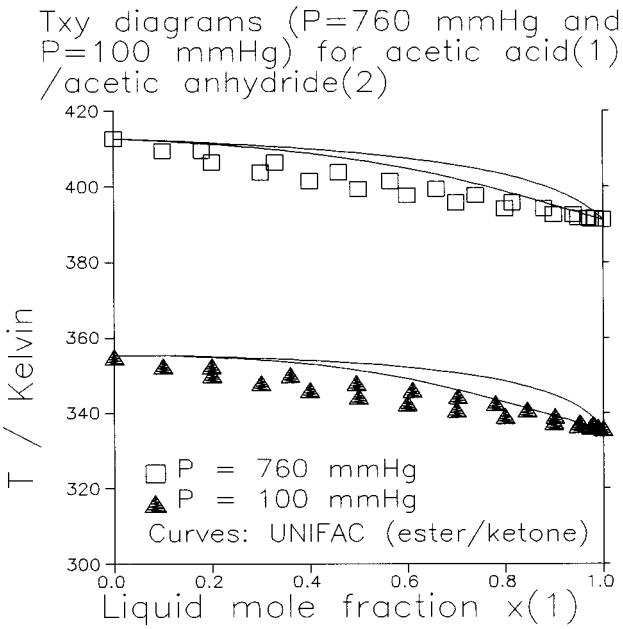


Figure 13. Txy diagrams[38] calculated with an ester/ketone combination. Compare with Figure 12.

should be a slightly positive correction, whereas in the other sets, a negative correction is needed. Therefore, the regression procedure yields some intermediate contribution. Fortunately, neither of the data are strongly nonideal. Yet, this observation indicates the kinds of difficulties that can arise when doing parameter regressions in series. It also suggests that the representation of ether mixtures, with the wide range of possible structural features exhibited by such systems, could benefit from additional group definitions, as has been done in the past, for example, for dioxanes.[5]

Anhydride interaction parameters are presented in Table 5. Examples of correlation results are given by Figures 11−18. Figure 11 compares the results for the cyclohexane/acetic anhydride binary[38] as calculated with the new parameters and the ester/ketone combination for the anhydride. In this case one may note the (false) Txy diagram or (false) liquid-phase instability which results when the ester/ketone combination is used. Figures 12 and 13 plot phase equilibria calculated for the system acetic acid/acetic anhydride[39] using the new groups and using an ester/ketone group combination. Figure 14 makes the same comparison in the case of the diisopropyl ether binary with acetic anhydride. Observations similar to those made in the case of Figure 11 are made here. Figures 15 and 16 show the results for benzene and methyl iodide binaries with acetic anhydride. Such systems, however, are not significantly improved by the definition of a new anhydride group.

Perhaps, of more interest are situations in which the parameters might fail. The problem of inaccuracy of group contribution methods for predicting activity coefficients of mixtures with proximity effects or polyfunctional species is well-known. There are data on a

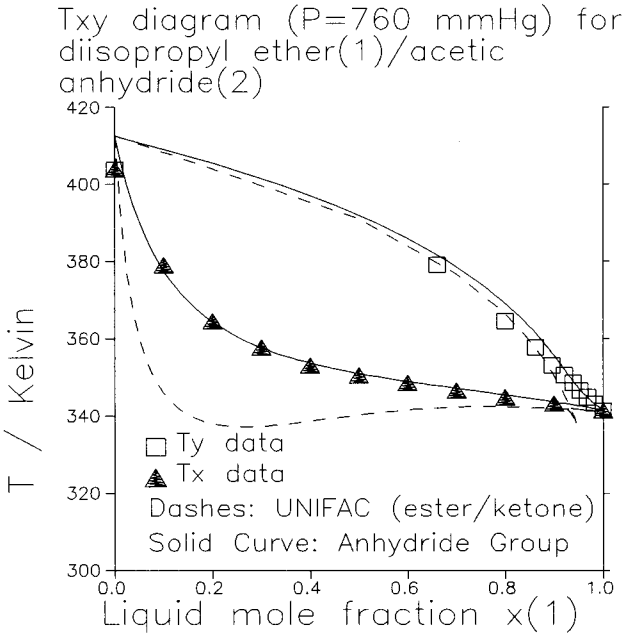


Figure 14. Diisopropyl ether/acetic anhydride Txy diagram[39] calculated with an ester/ketone combination and with new anhydride parameters.

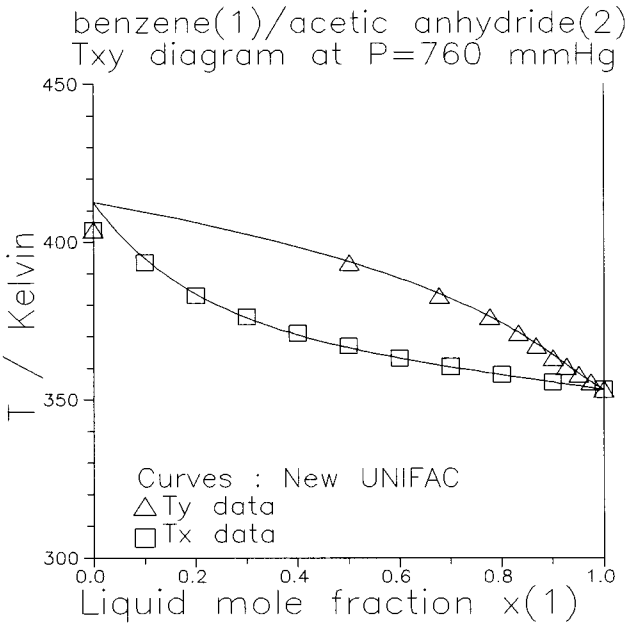


Figure 15. Benzene/acetic anhydride Txy diagram.[40]

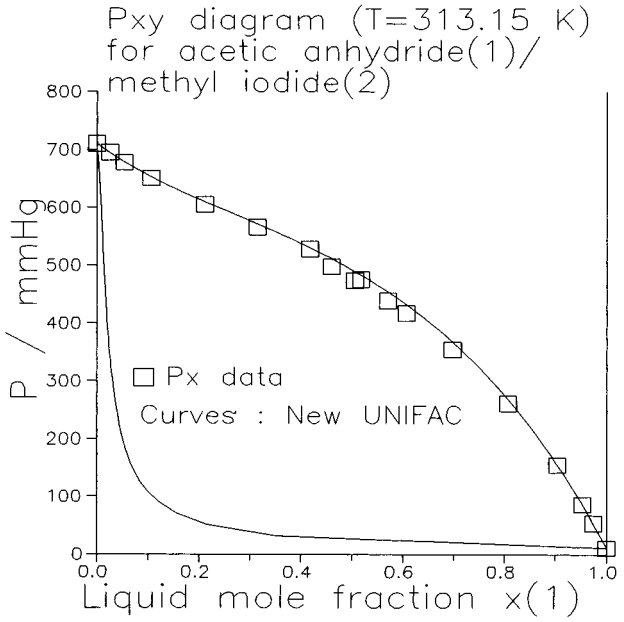


Figure 16. Acetic anhydride (1)/methyl iodide (2) Pxy diagram.[41]

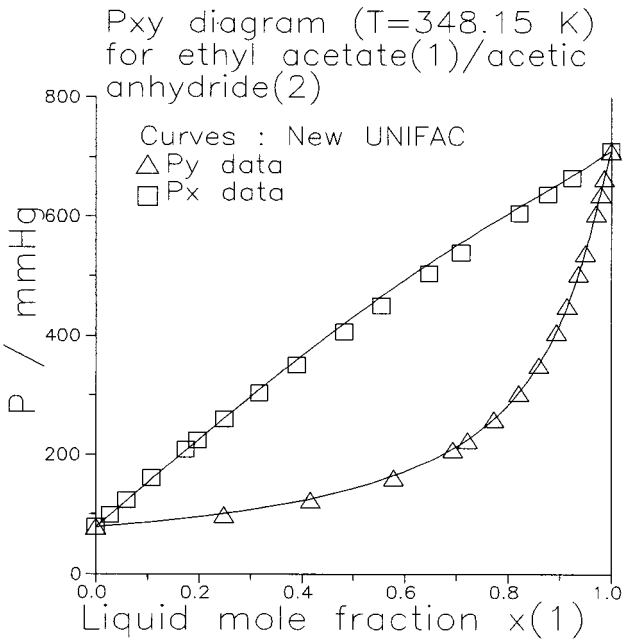


Figure 17. Ethyl acetate/acetic anhydride Pxy diagram.[42]

number of ester-containing chemicals with anhydrides: both monofunctional esters and esters with more than a single functional group, e.g., allyl acetate, methylene diacetate, and isopropenyl acetate. We have emphasized representing data on the monofunctional esters such as methyl and ethyl acetate. Figure 17 shows correlation results for ethyl acetate with acetic anhydride, a system exhibiting weak positive deviation. Figure 18, on the other hand, shows results for methylene diacetate with acetic anhydride. This system deviates slightly negatively from Raoult's law. In Figure 18 the results are reasonable, but this might not have been the case had the differences in ideality been larger. Representation of allyl acetate is also reasonably good with the present parameters, while data on isopropenyl acetate are not satisfactorily correlated with the parameters in Table 5. Thus, polyfunctional systems may occasionally be well-represented as in the case of methylene diacetate. Yet, in practice, great caution remains the general rule in such cases.

Heats of mixing are harder to correlate. Often the temperature-independent parameters in local composition G^E models are simply not adequate to predict the experimental H^E. A quite common case shows a positive G^E and a positive H^E which is underpredicted by differentiation of the G^E model and with a serious risk of predicting false liquid instability if one tries to match the experimental H^E by data reduction. Whenever we could check our parameters against H^E data, the sign of the heat effect was normally correct but the magnitude was only occasionally correct. This means the temperature dependence of the activity coefficients calculated with the new parameters should be reasonable but not quantitative over wide temperature ranges.

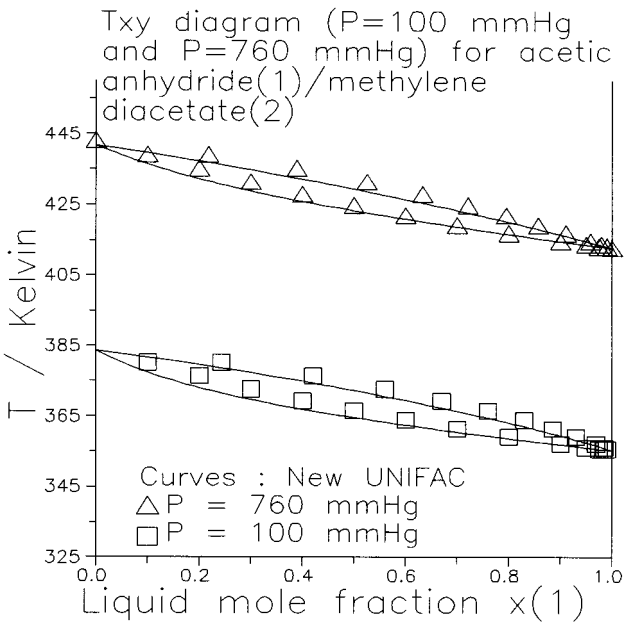


Figure 18. Acetic anhydride/methylene diacetate Txy diagrams.[38]

It is well-known[4,5] that it takes temperature-dependent model parameters with at least two (sometimes three) terms to correlate the order of magnitude of heats of mixing in strongly temperature-dependent systems. Yet, even with such elaborate equations, the concentration dependence of H^E, which is often dramatically much more complex than that of G^E, is still difficult to correlate in many cases.

Discussion

All results shown and all parameters presented are associated with a UNIFAC model with temperature-independent parameters. Parameters for such models might seem obsolete when one considers the proven advantages of temperature-dependent parameters for strongly temperature-dependent systems and the general developments over the past decade in applications of sophisticated liquid solution theories to prediction of phase equilibria.[43]

It is undoubtedly true that results obtained with temperature-dependent parameters should be more accurate if the model parameters are based on a sufficient amount of reliable temperature derivative (first and second) data. For the present systems and for several other systems not presently covered by UNIFAC models, there is not plenty of such reliable data in the open literature yet. Thus, there may still be some incentive to treat such cases with the simpler original UNIFAC. The parameters of this paper should be subject to later revisions should new and better data appear. In the meantime, however, the present parameters may provide useful estimates of phase equilibria. More sophisticated recent approaches[43] may perhaps be developed to application ranges as wide as or maybe even wider than solution-of-groups methods. Still, from a practical and scientific point of view, we regard it as important and scopeful to address cases when useful results can be found with simple and yet quite general approaches.

A basic premise in the solution-of-groups method is, of course, that the combinatorial term can adequately predict property variations unaffected by interaction effects. On the basis of this observation, we[44] studied characteristic chain-length dependencies and other structural relationships imbedded in the model. That study showed that the combinatorial term has subtle but major implications on the correlation of activity coefficients and derivatives. Thus, while current models may be generally accurate, there are a number of cases where they will be incorrect either because the combinatorial terms give wrong dependencies of properties on the number of groups in solute/solvent molecules or because they make extremely large contributions to the total excess properties, perhaps constraining residual contributions to unrealistic values. Nevertheless, we emphasize the potential utility of this contribution. The parameters given here should be particularly useful for developing integrated computer-aided process engineering tools, for solvent design and other traditional engineering calculations.

Summary

Group definitions are determined together with structural parameters for a set of subgroups within four different main groups. Numerous data are collected and used for determining interaction parameter values. Selected results of the correlation are presented and discussed.

Acknowledgment

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Appendix A. UNIFAC Equations

The activity coefficient of species i in solution is composed of two terms: a combinatorial term and a residual term. The combinatorial term can be written as

$$\ln \gamma_i^C = 1 - J_i + \ln J_i - \frac{z}{2} q_i \left(1 - \frac{J_i}{L_i} + \ln \frac{J_i}{L_i}\right) \quad \text{(A.1)}$$

with

$$L_i = \frac{q_i}{\sum_j x_j q_j}; \quad J_i = \frac{r_i}{\sum_j x_j r_j} \quad \text{(A.2)}$$

where the sums over j run over all components in the mixture. The residual term can be written as

$$\ln \gamma_i = q_i(1 - \ln L_i) - \sum_k \left(\frac{s_{ki}}{\eta_k}\vartheta_k - G_{ki} \ln \frac{s_{ki}}{\eta_k}\right) \quad \text{(A.3)}$$

where the sum over k runs over all main groups in solution. In eq A.3

$$G_{ki} = \sum_{l \in k} \nu_{li} Q_l; \quad \vartheta_k = \sum_i x_i G_{ki} \quad \text{(A.4)}$$

and

$$s_{ki} = \sum_m G_{mi} \tau_{mk}; \quad \eta_k = \sum_i x_i s_{ki} \quad \text{(A.5)}$$

In eqs A.4 and A.5, l runs over all subgroups in main group k and m runs over all main groups in solution. Finally

$$\tau_{mn} = \exp(-a_{mn}/T) \quad \text{(A.6)}$$

where m and n denote main groups.

Appendix B. Geometrical Parameters

This appendix summarizes the motivation behind the $\boldsymbol{R}$ and $\boldsymbol{Q}$ values listed in Table 1.

AC−Br. Values exist[18] in the case of AC−Br, namely, $V_{wk} = 20.66\ \text{cm}^3\ \text{mol}^{-1}$ and $A_{wk} = 2.43 \times 10^9\ \text{cm}^2\ \text{mol}^{-1}$. These were normalized using the equations

$$R_{\text{AC-Br}} = 20.66/15.17 = 1.3619 \quad \text{(B.1)}$$

$$Q_{\text{AC-Br}} = 2.43 \times 10^9/2.5 \times 10^9 = 0.9720 \quad \text{(B.2)}$$

and inserted in Table 1.

AC−CN. As in the preceding case, values exist.[18] These were normalized as above

$$R_{\text{AC-CN}} = 20.24/15.17 = 1.3342 \quad \text{(B.3)}$$

$$Q_{\text{AC-CN}} = 2.49 \times 10^9/2.5 \times 10^9 = 0.9960 \quad \text{(B.4)}$$

and inserted in Table 1.

Epoxy. Our definition of the epoxy subgroup follows that of previous treatments,[6] but a few subgroups are added for flexibility. Previously, $\boldsymbol{R}$ and $\boldsymbol{Q}$ values were given for CH_2OCH- and for a $-CHOCH-$ subgroup. These are implemented in this work together with their interaction parameters with CH_2, C═C, and CCl. $\boldsymbol{R}$ and $\boldsymbol{Q}$ values for other subgroups within the epoxy main group were obtained as follows:

$$R_{\text{CH}_2\text{OCH}_2} = R_{\text{CH}_2\text{OCH}} + R_{\text{CH}_2} - R_{\text{CH}} = 1.3652 + 0.6744 - 0.4469 = 1.5927 \quad \text{(B.5)}$$

$$Q_{\text{CH}_2\text{OCH}_2} = Q_{\text{CH}_2\text{OCH}} + Q_{\text{CH}_2} - Q_{\text{CH}} = 1.0080 + 0.5400 - 0.2280 = 1.320 \quad \text{(B.6)}$$

and

$$R_{\text{CH}_2\text{OC}} = R_{\text{CH}_2\text{OCH}} + R_{\text{C}} - R_{\text{CH}} = 1.3652 + 0.2195 - 0.4469 = 1.1378 \quad \text{(B.7)}$$

$$Q_{\text{CH}_2\text{OC}} = Q_{\text{CH}_2\text{OCH}} + Q_{\text{C}} - Q_{\text{CH}} = 1.0080 + 0.0000 - 0.2280 = 0.7800 \quad \text{(B.8)}$$

and

$$R_{\text{CHOC}} = R_{\text{CH}_2\text{OCH}} + R_{\text{C}} - R_{\text{CH}_2} = 1.3652 + 0.2195 - 0.6744 = 0.9103 \quad \text{(B.9)}$$

$$Q_{\text{CHOC}} = Q_{\text{CH}_2\text{OCH}} + Q_{\text{C}} - Q_{\text{CH}_2} = 1.3652 + 0.2195 - 0.5400 = 0.4680 \quad \text{(B.10)}$$

and finally

$$R_{\text{COC}} = R_{\text{CHOC}} + R_{\text{C}} - R_{\text{CH}} = 0.9103 + 0.2195 - 0.4469 = 0.6829 \quad \text{(B.11)}$$

$$Q_{\text{COC}} = Q_{\text{CHOC}} + Q_{\text{C}} - Q_{\text{CH}} = 0.4680 + 0.0000 - 0.2280 = 0.2400 \quad \text{(B.12)}$$

Gmehling and co-workers[12] give different $\boldsymbol{R}$ and $\boldsymbol{Q}$ values for $-CHOCH-$ and CH_2OCH-. They give no values for other subgroups.

Anhydrides. No values are directly available for anhydride subgroups. Values may be determined by combining contributions of other subgroups. There is more than one way of doing that: One possibility is to construct the anhydride group, O═COC═O, from two ketones ("═"C═O) and a single ether ("═"−O−) atom. That would give the values for O═COC═O:

$$R_{\text{O=COC=O}} = 2R_{\text{C=O}} + R_{\text{-O-}} = 2(0.7713) + 0.2439 = 1.7865 \quad \text{(B.13)}$$

$$Q_{\text{O=COC=O}} = 2Q_{\text{C=O}} + Q_{\text{-O-}} = 2(0.6400) + 0.2400 = 1.5200 \quad \text{(B.14)}$$

Alternatively, the anhydride group, O═COC═O, might be considered as a combination of an ester ("═"−COO) and a ketone ("═"C═O). That would give

$$R_{\text{O=COC=O}} = R_{\text{C=O}} + R_{\text{-COO}} = 0.7713 + 1.002 = 1.7732 \quad \text{(B.15)}$$

$$Q_{\text{O=COC=O}} = Q_{\text{C=O}} + Q_{\text{-COO}} = 0.6400 + 0.88 = 1.5200 \quad \text{(B.16)}$$

These two results are not very different. In our calculations, the latter results as given by eqs B.15 and B.16 were taken, as shown in Table 1. Anhydrides also exist as mixed anhydrides, that is, anhydrides with two different fatty acid radicals. Most of these substances can be constructed from the groups given above, except acetoformic anhydride. We have not made any calculations for formic anhydride systems, and it is likely that their properties might be significantly different from those of higher anhydrides. If such systems were to be pursued, their $\boldsymbol{R}$ and $\boldsymbol{Q}$ values could readily be determined in an analogous fashion. For example, one could construct the O═CHOC═O subgroup as follows:

$$R_{\text{O=CHOC=O}} = R_{\text{CHO}} + R_{\text{C=O}} + R_{\text{-O-}} \quad \text{(B.17)}$$

$$Q_{\text{O=CHOC=O}} = Q_{\text{CHO}} + Q_{\text{C=O}} + Q_{\text{-O-}} \quad \text{(B.18)}$$

That would give

$$R_{\text{O=CHOC=O}} = 0.9980 + 0.7713 + 0.2439 = 2.0132 \quad \text{(B.19)}$$

$$Q_{\text{O=CHOC=O}} = 0.9480 + 0.6400 + 0.2400 = 1.8280 \quad \text{(B.20)}$$

Similarly, for O═CHOHC═O:

$$R_{\text{O=CHOCH=O}} = 2R_{\text{CHO}} + R_{\text{-O-}} = 2(0.9980) + 0.2439 = 2.2399 \quad \text{(B.21)}$$

$$Q_{\text{O=CHOCH=O}} = 2Q_{\text{CHO}} + Q_{\text{-O-}} = 2(0.9480) + 0.2400 = 2.1360 \quad \text{(B.22)}$$

List of Symbols

A_{wk} = molecular surface area (cm^2/mol) of group k, eq 1
a_{mn} = interaction energy parameter of local composition models, eq A.6
f_i = sum of squared deviations (model/measurements) including data of type i

G = Gibbs free energy
G_{ki} = defined by eq A.4
H = enthalpy
J_i = defined by eq A.2
L_i = defined by eq A.2
r_i = volume parameter of component i
q_i = surface area parameter of species i
Q_k = surface area parameter of group k
R_k = volume parameter of group k
s_{ki} = defined by eq A.5
T = temperature
V_{wk} = molecular volume (cm^3/mol) of group k, eq 1
x_i = mole fraction of species i in solution
x_{ij} = mole fraction solubility of solute i in solvent j

Greek Letters

α = regularization factor, eq 8
γ_{ij} = activity coefficient of solute i in solvent j
δ = regularization parameter (eq 8), typically around 10^5
ν_{ki} = number of groups of type k in component i
τ_{ij} = local composition weighting factors, defined by eq A.6
η_k = defined by eq A.5
ϑ_m = defined by eq A.4
σ_i = weighting factor of property i

Subscript

i = species i

Superscripts

C = combinatorial property
E = excess property
R = residual property
∞ = at infinite dilution

Abbreviations

LLE = liquid liquid equilibrium
SLE = solid liquid equilibrium
VLE = vapor liquid equilibrium

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